MIL-C-47247(MI)
26 July 1974
SUPERSEDING
MIS-14003
9 October 1972

MILITARY SPECIFICATION

COATING, PROTECTIVE, RESIN, SILICONE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of a rigid one-component silicone resin for use as a protective coating for the surfaces and junctions of transistors, diodes, and other electronic devices.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on the date of the invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on the date of invitation for bids or request for proposal shall apply.

FSC 8030

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American Society for Testing and Materials

ASTM D 445	Viscosity of Transparent and Opaque Liquids (Kinematic and Dynamic Viscosities)
ASTM D 1298	Specific Gravity of Petroleum Liquids
ASTM D 1346	Testing Electrical Insulating Varnishes for 180°C and above
ASTM D 1544	Color of Transparent Liquids (Gardner color Scale)

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Uniform Classification Committee

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois, 60605)

National Motor Freight Traffic Associations, Inc.
National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc. 1616 P Street, N.W., Washington, D.C. 20036)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet all the requirements of this specification.

3.2 Materials.

3.2.1 Composition and condition in container.

3.2.1.1 Composition. The materials shall be a silicone resin compound dispersed in an aromatic solvent.

3.2.1.2 Condition in container. The material in the container shal be clear and free of gel particles and foreign matter that would adversely affect the intended purpose.

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3.2.2 Shelf life. The resin, as furnished by the manufacturer, shall be capable of meeting the requirements of this specification after 6 months storage (from date of shipment) in the original unopened containers at a temperature not exceeding 26.7 degrees Celsius (C) (80 degrees Fahrenheit (F)).

3.3 Physical properties of uncured (as supplied) resin. The uncured resin shall meet the physical properties listed in Table I.

TABLE I

PROPERTIES OF UNCURED RESIN	
Characteristic	Requirement
Color, Gardner	1 maximum
Viscosity at 25°C (Centipoise)	80 - 140
(Non-volatile content (percent) (1-1/2 gms/3hrs./135°C)	49-52

3.4 Workmanship. The resin covered by this specification shall be homogeneous and free of foreign materials. Containers shall be clean, uniformly filled, well-sealed, and legibly marked.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. Unless otherwise specified in the contractual document (see 6.2), a sample consisting of not less than one pound of resin, manufactured under the same conditions as those proposed for subsequent production, shall be subjected to preproduction inspection. Accomplishment of preproduction inspection shall be as specified herein. If the preproduction sample does not meet the requirements of this specification, it shall be rejected. Subsequent units shall not be considered for acceptance until Government approval of the preproduction sample has been obtained. Units subjected to preproduction inspection shall have successfully passed acceptance inspection. Testing of the preproduction sample, to determine compliance with the characteristics listed in Table II, shall be conducted in accordance with the corresponding test paragraphs.

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TABLE II

PREPRODUCTION INSPECTION		
Characteristic	Requirement source	Test paragraph
Composition and condition in container	3.2.1	4.5.3.1
Color, Gardner	Table I	4.5.3.2
Viscosity	Table I	4.5.3.3
% Non-volatile content	Table I	4.5.3.4
Shelf life	3.2.2	4.5.3.5
Workmanship	3.4	4.5.3.6

4.3 Quality conformance inspection.

4.3.1 Sampling for acceptance inspction. Sampling for acceptance inspection shall be conducted in accordance with MIL-STD-105, at an acceptable quality level (AQL) of 4.0 percent defective.

4.3.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.3.3 Acceptance inspection. Acceptance inspection of the sample specified in 4.3.1, to determine compliance with the characteristics specified in Table III, shall be conducted in accordance with the corresponding test and inspection paragraphs.

TABLE III

ACCEPTANCE INSPECTION		
Characteristic	Requirement source	Test and inspection paragraph
Condition in container	3.2.1.2	4.5.3.1
Workmanship	3.4	4.7
Packaging, packing and marking	5.	4.6

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4.4 Inspection equipment. Unless otherwise specified (see 6.3), the inspection equipment for conducting examination and tests shall be as specified in the applicable test methods and procedure paragraphs.

4.5 Test methods procedures.

4.5.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements:

a. Temperature, room ambient (16 to 32 degrees C (60 to 90 degrees F)).

b. Altitude, facility ground.

c. Humidity, facility ambient up to 95 percent relative humidity.

4.5.2 Test sequence. Testing sequence within each classification of inspection (preproduction or acceptance) shall be at the option of the supplier.

4.5.3 Tests.

4.5.3.1 Composition and condition in container. Unless otherwise specified, composition shall be certified by the supplier. The condition in the container shall be visually examined to establish compliance with 3.2.1.

4.5.3.2 Color, gardner. Color shall be determined in accordance with ASTM D-1544 and shall conform to 3.3.

4.5.3.3 Viscosity. The viscosity of the resin as supplied shall be determined in accordance with ASTM D-445 using an Ubbelohde tube, and shall conform to 3.3.

4.5.3.4 Non-volatile content. The percent non-volatile content shall be determined in accordance with ASTM D-1346 (1-1/2 gms/3hrs./135 degrees C) and shall conform to 3.3.

4.5.3.5 Shelf life. The shelf life shall be certified by the manufacturer and shall conform to 3.2.2 on the date of shipment.

4.6 Packaging, packing, and marking. The inspector shall ascertain that the packaging, packing, and marking of the material conforms to this specification.

4.7 Workmanship. The material and containers shall be visually examined for compliance with 3.4.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. The material furnished under this specification shall be packaged in a manner assuring no damage during shipping, handling, and storage.

5.2 Marking. Each package and exterior shipping container shall be clearly and legibly marked. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name and address.

b. Manufacturer's designation.

c. Lot or batch number.

d. Date of manufacturer or shipment (see 3.2.2).

e. Number of this specification.

5.2.1 Precautionary marking. In addition to the above, all packages and exterior shipping containers shall be marked with the following precautionary marking:

WARNING: FLAMMABLE Keep away from heat and flame. Use with adequate ventilation. Avoid prolonged breathing of vapor. Avoid prolonged or repeated skin contact.

6. NOTES

6.1 Intended use. The silicone resin covered by this specification is intended for use as a rigid protective coating for the surfaces and junctions of transistors, diodes and other electronic devices. It may be thinned with electronic grade toluene or Xylene.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, numbe, and date of this specification.

b. Whether a preproduction sample is required (see 3.1) and if so, pertinent details.

c. Requirement for certification of composition (see 4.5.3.1).

6.3 Definitions.

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6.3.1 Typical electrical properties:

a. Dielectric constant at 100 Hz, 3.22; at 1000KHz, 3.15

b. Dissipation factor at 100 Hz, 0.008; at 1000KHz, 0.004

c. Dielectric strength 1800 volts/mil

d. Volume resistivity 26 ohm-cm X 10^{15}

6.3.2 Flash point. 80 degrees minimum.

6.4 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-14003, dated 9 October 1972.

Custodian:
ARMY-MI

Preparing Activity:
ARMY - MI
Project No. 8030-A064

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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